Racial Inequity in Respiratory Care Across Northeast Kansas

DataVision Avengers

ASA DataFest 2026

Research Overview

An analysis of 390,732 respiratory encounters at Stormont Vail Health (2022–2025) revealing how Black patients face systemic barriers in respiratory care access, and actionable recommendations for Stormont Vail Health (SVH) to mitigate this issue in their service area.

The National Crisis

- **25.7 Million** Americans currently suffer from asthma.
- Black Americans are **1.5x more likely** to have an asthma diagnosis.
- Black Americans are **5x more likely** to visit the Emergency Department (ED) for asthma.
- Black Americans are **3x more likely** to die from asthma complications.
- **Pediatric Impact:** Black Medicaid-enrolled children have **2.42x the odds** of an asthma ED visit compared to White children.

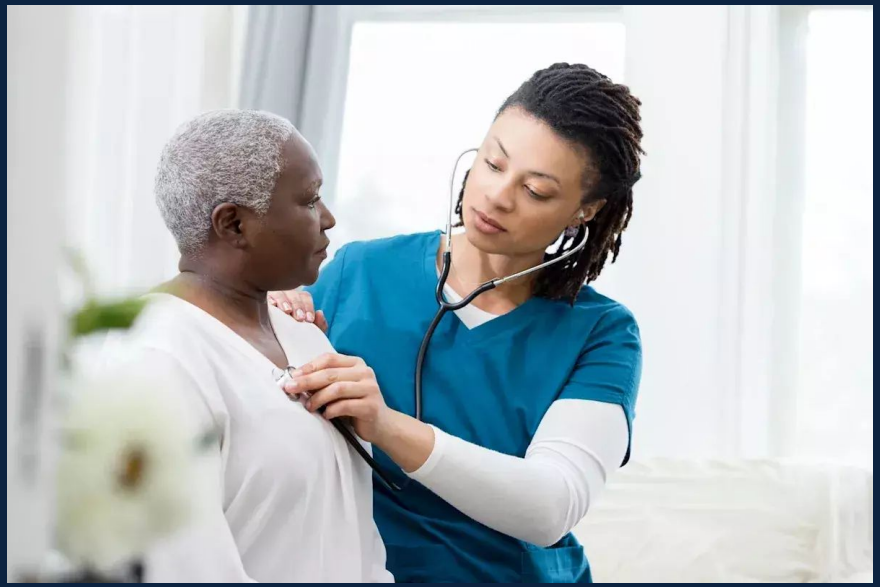


Image: LIV Hospital

The Kansas Context



Image: TK Business Magazine

Medicaid Expansion: Kansas is 1 of 9 US states that has NOT expanded Medicaid.

Coverage Gap: ~39,000 adults earn too much for Medicaid but too little for ACA subsidies.

Legislative Block: Expansion has been proposed 7 times (2019-2025) and blocked every time.

Regional Precedent: In the KC metro, Black residents are 14.9% of the population but account for **47.4%** of asthma ED visits.

Merged with SDOH screening, provider data, and ACS socioeconomics.

Our Dataset: Stormont Vail Health (2022-2025)

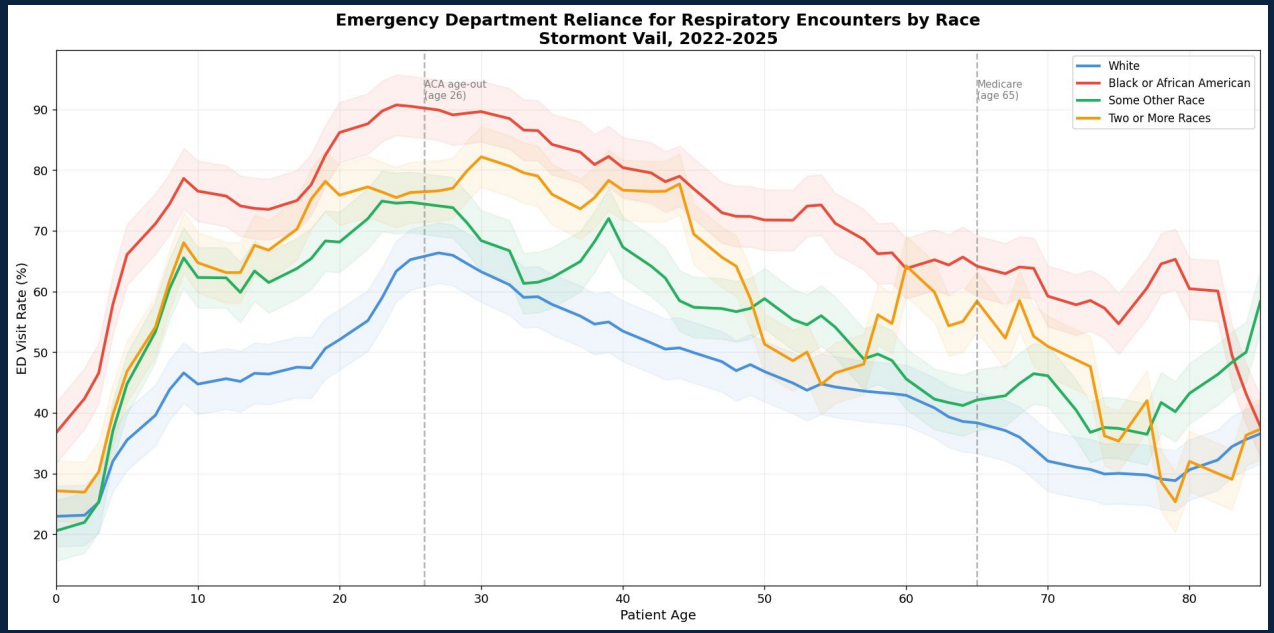
7,675,801
Total Encounters
Comprehensive system-wide visits across all departments.

390,732
Respiratory Encounters
Identified via ICD-10 J-codes, U07 (COVID), and R05-R06.

33, 771
Unique Patients

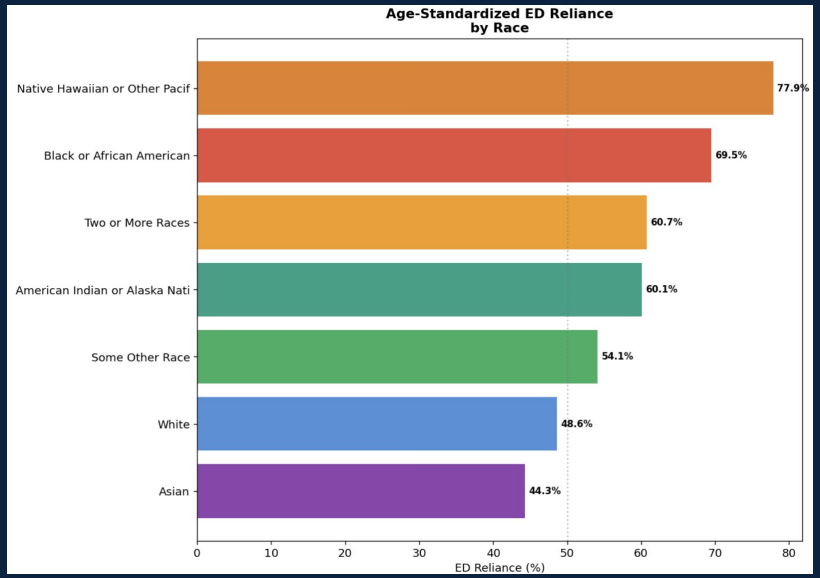
Data Linkage
Holistic Context

Our Signal: ED Reliance for Respiratory Encounters by Race

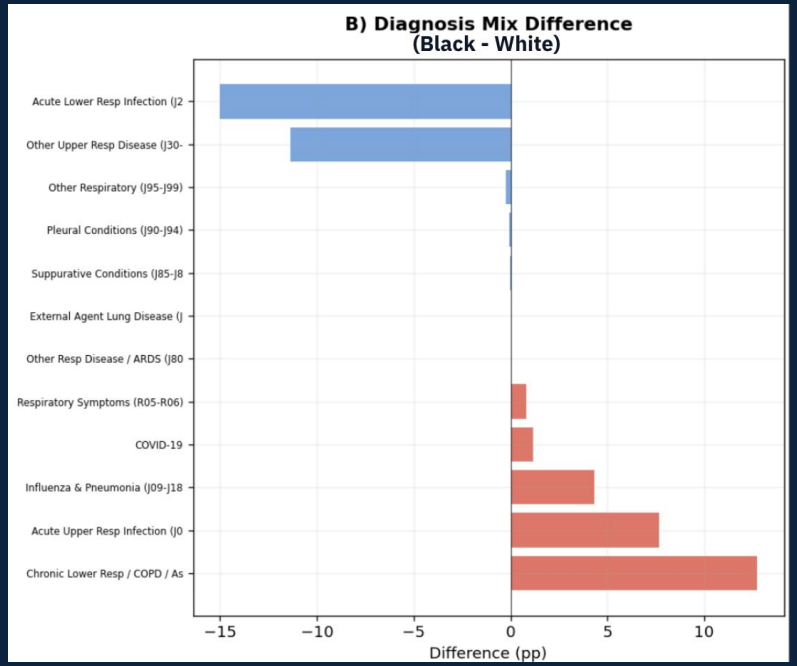


Key Takeaway: The disparity persists from infancy through old age, remaining remarkably unaffected by major healthcare policy thresholds like ACA dependency limits (age 26) or medicare eligibility (age 65)

Our Signal: ED Reliance Index + Diagnosis Mix



Age Standardized ED Reliance Index: Proportion of care received in the ED vs. Outpatient settings. A high score indicates the population relies primarily on emergency care rather than disease management.

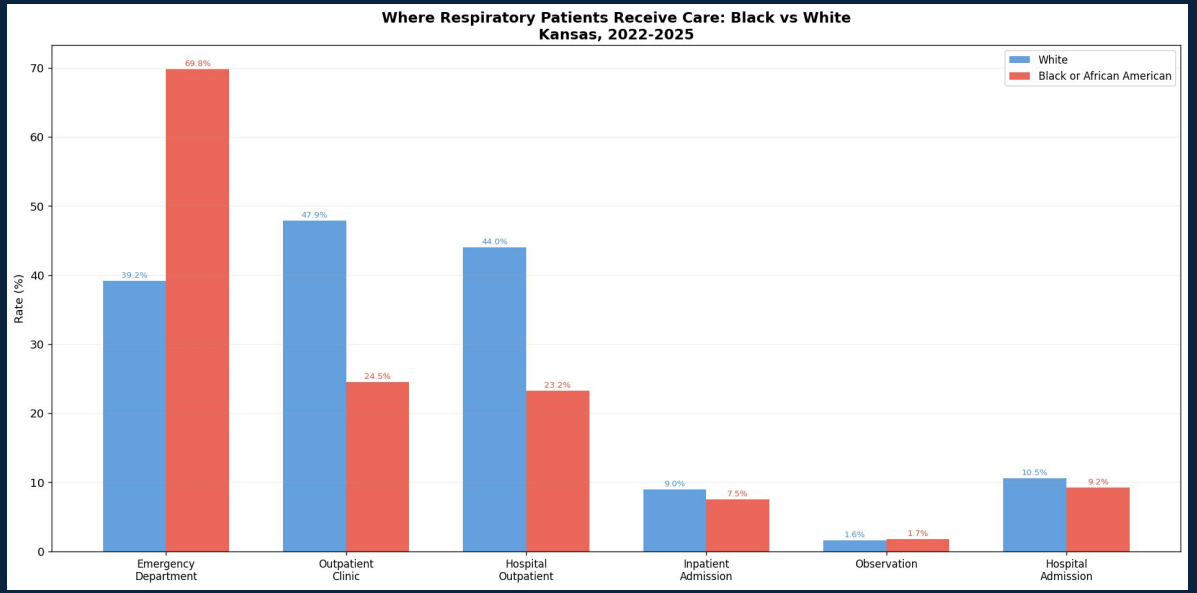


Diagnosis Mix Difference: Percentage point difference in diagnosis. Blue conditions are diagnosed more often for White and Red conditions are diagnosed more often for Black

Point of Care Analysis

Black patients in our dataset have the highest per-capita rate of asthma and COPD - meant to be managed in outpatient clinics with inhalers and action plans.

"point of care" analysis reveals that the healthcare system is utilizing the Emergency Department as a substitute for primary care for Black patients.



Ruling Out Confounders: Structural Stability

To ensure this disparity isn't an artifact of behavior or environment, we controlled for critical variables. The gap mathematically survives every stratification.



Smoking Status

The gap remains identical in magnitude across **never-smokers, former smokers, and current smokers.** Behavioral choices do not explain the structural access deficit.



Seasonality

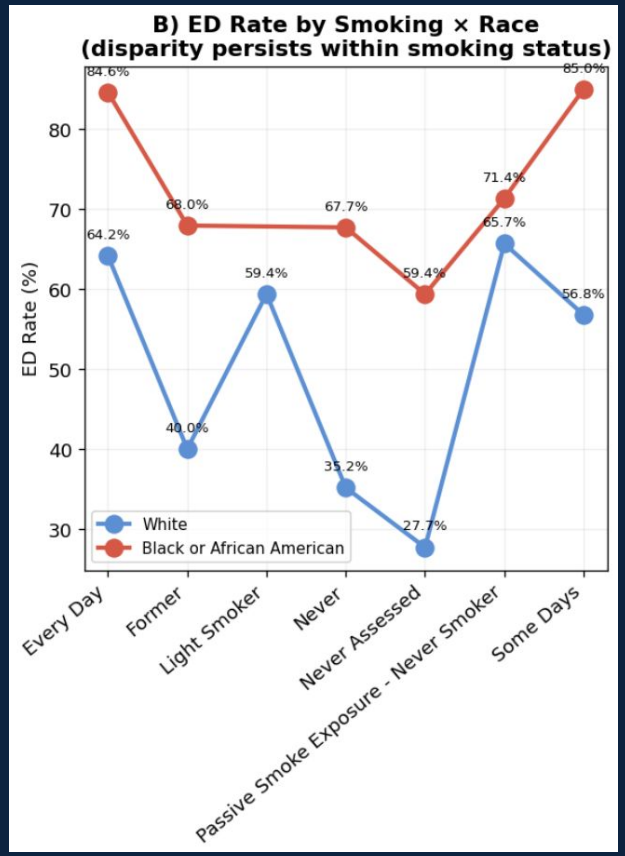
The disparity persists symmetrically across **all four seasons**, ruling out acute environmental triggers (like pollen blooms or winter viruses) as the primary differentiator.



Temporal Trends

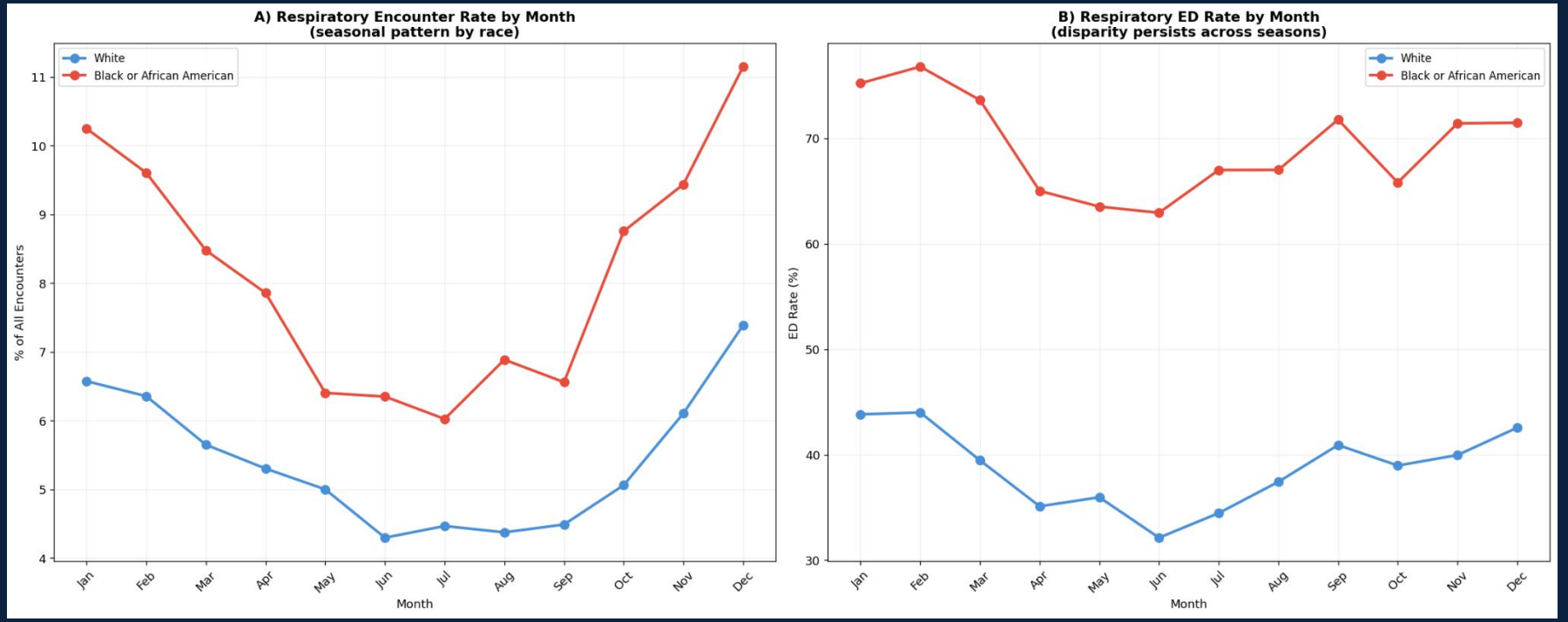
Analyzing year-over-year data from **2022 to 2025** shows a flatline. There has been no meaningful improvement or convergence in the access gap over three years.

Confounder Analysis: Smoking Status

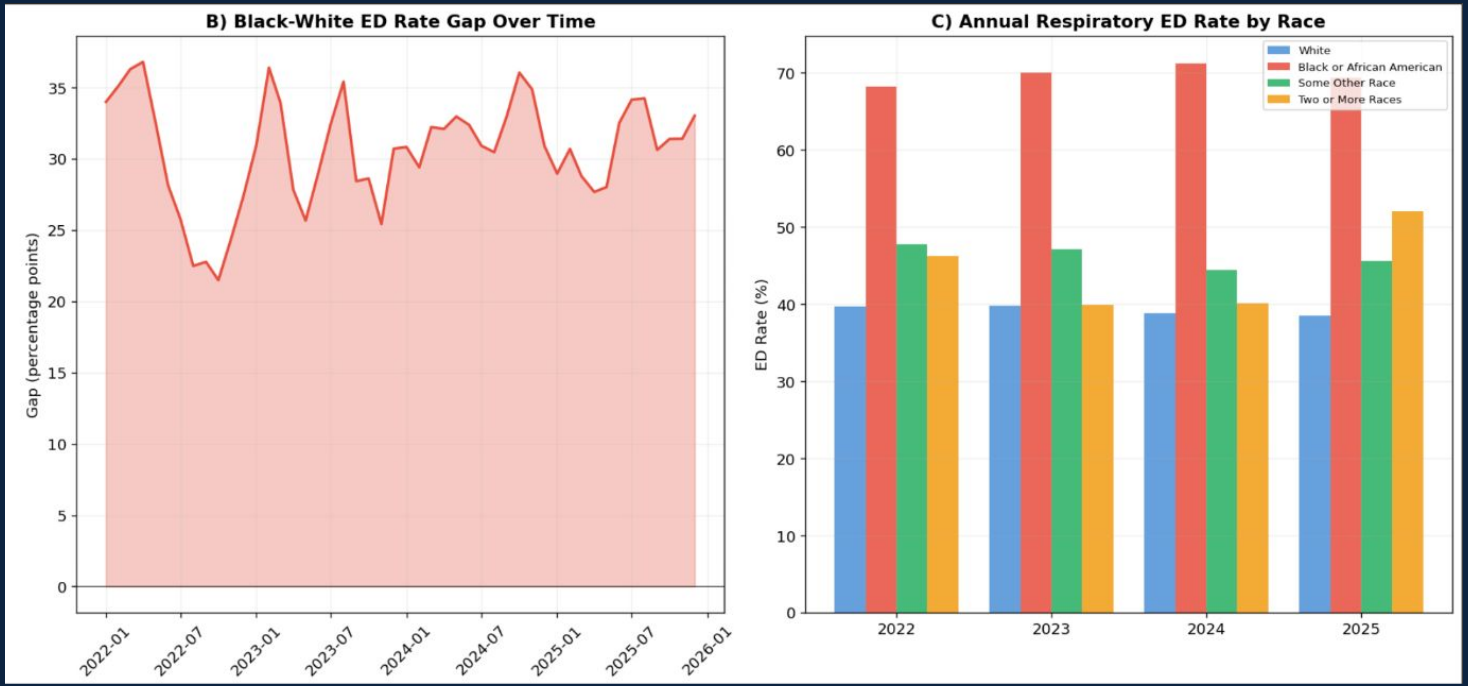


Metric	Value	Interpretation
CMH Statistic	**1537.69**	Extremely strong association.
p-value	**< 0.001**	Statistically significant ($YES\checkmark$).
Common Odds Ratio	**3.657**	**3.66x** higher odds for Black patients.

Confounder Analysis: Seasonality

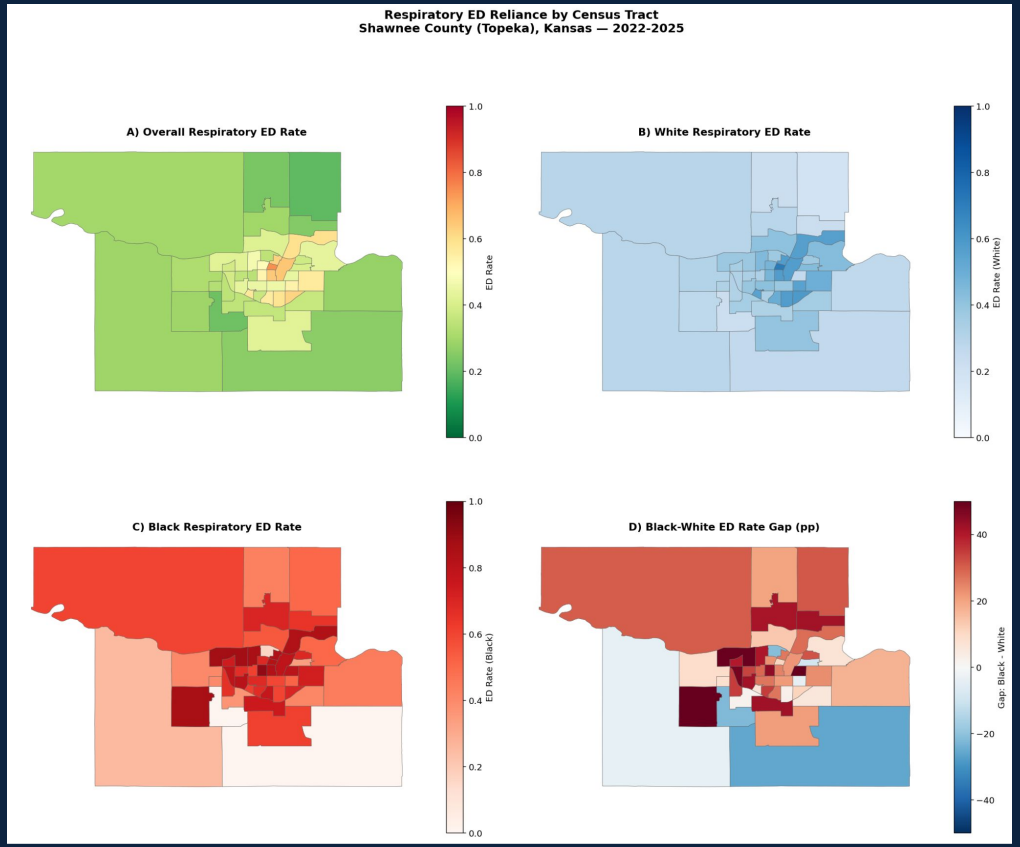


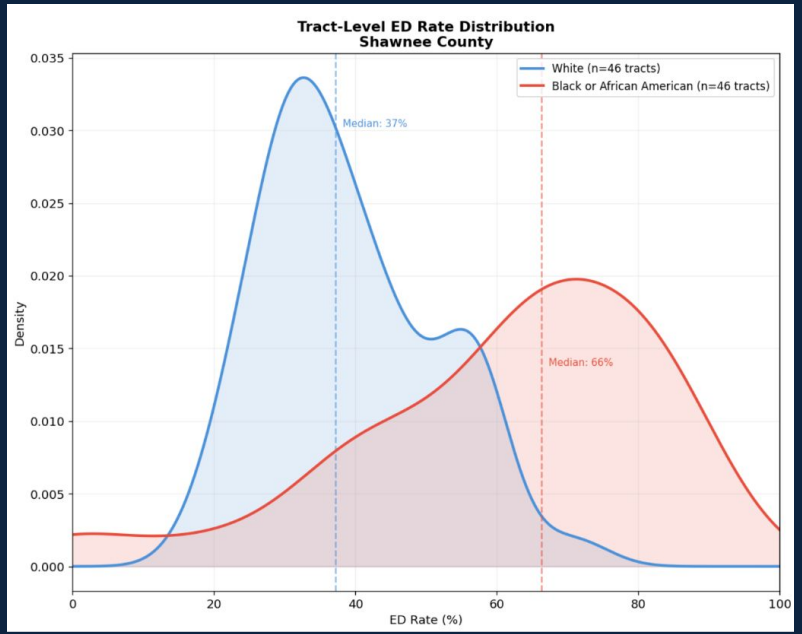
Confounder Analysis: Temporal Trends



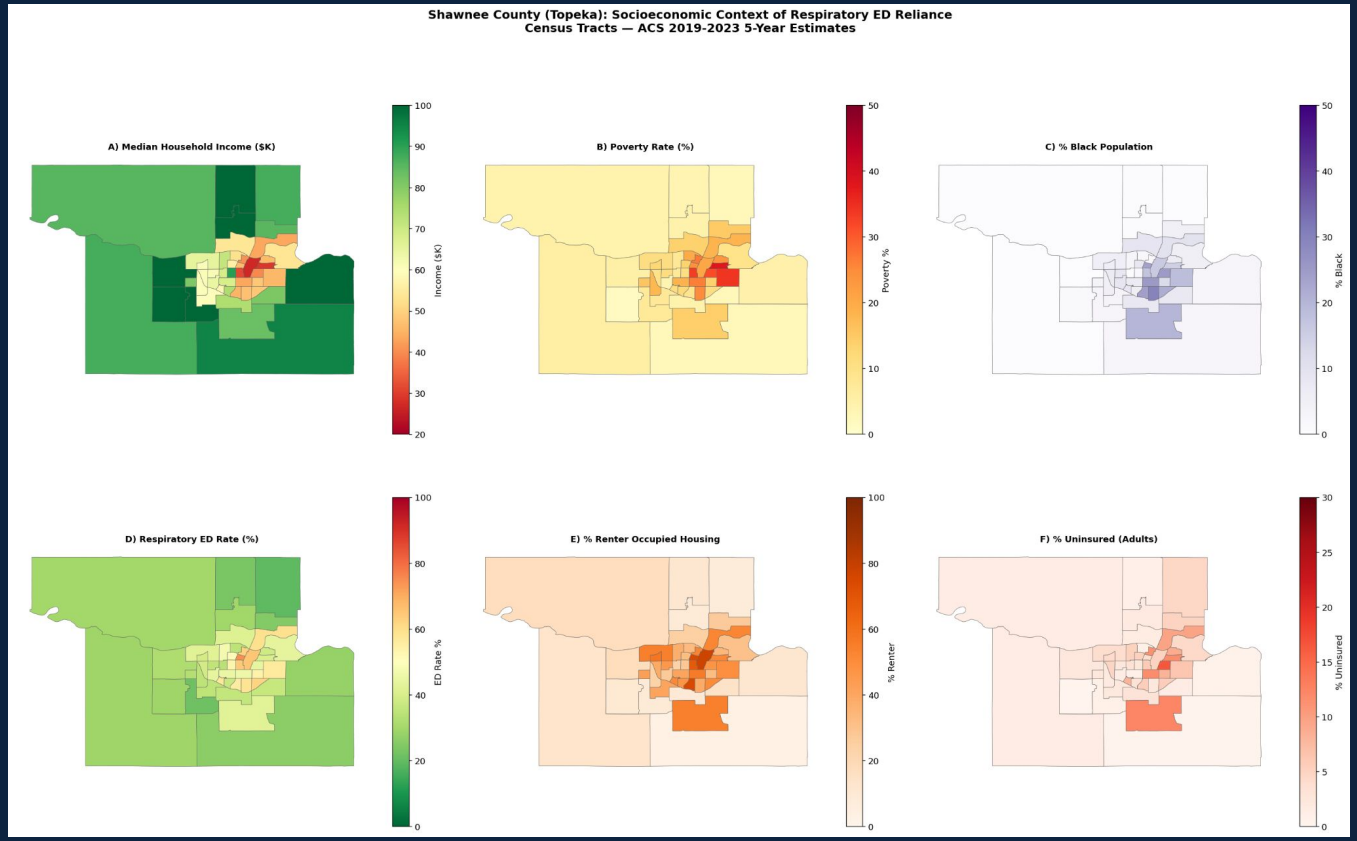
Key Takeaway: The gap is not closing, it has stayed relatively stable indicating no real improvement

Geographical Data Reinforces Disparity





Geographical Data: Socioeconomic Context



Our Key Findings

The Core Reality: Black patients rely on Emergency Department (ED) care at significantly higher rates than White patients—a gap that remains stagnant and structural.

- **The Persistence of the Gap:**
  - **Black Patients:** 60–80% ED reliance across **all** age groups.
  - **White Patients:** 25–60% ED reliance (dropping below 50% by age 30).
- **A Structural, Not Clinical, Issue:** Disparities persist even after controlling for:
  - Smoking status & Diagnosis (Asthma/COPD).
  - Seasonality & Age stratification.
- **Critical Concern:** Pediatric asthma demonstrates the widest, most immediate disparity.
- **Trend:** Zero improvement recorded between **2022 and 2025**.

Data-Driven Recommendations

Health Systems	Policymakers	Providers
Expand Outpatient Access: Target areas with high Black ED reliance.	**Medicaid Expansion:** Close the gap for ~39k Kansans (est. 5.5% ED reduction).	**Standardize Discharge:** Ensure Asthma Action Plans for *all* patients.
Community Health: Deploy home management for pediatric asthma.	**Environmental Justice:** Target housing and air quality in segregated areas.	**Follow-up Continuity:** Book outpatient appointments *before* ED discharge.
After-Hours Care: Divert non-emergent visits to dedicated clinics.	**Funding:** Support SDOH screening and mitigation programs.	**Bias Mitigation:** Direct training on severity assessment equity.

References

- Smith LB, O'Brien C, Kenney GM, Waidmann TA. Black–White Disparities in Asthma Hospitalizations and ED Visits Among Medicaid-Enrolled Children. *Hospital Pediatrics*. 2024;14(6):490-498. doi:10.1542/hpeds.2023-007477 — *Used for:* National benchmark that Black Medicaid children have 2.42x the asthma ED rate of White children, even after adjusting for sex, age, state, rurality
- Binney S, Flanders WD, Sircar K, Idubor O. Trends in US Pediatric Asthma Hospitalizations, by Race and Ethnicity, 2012–2020. *Preventing Chronic Disease*. 2024;21:240049. doi:10.5888/pcd21.240049 — *Used for:* National evidence that pediatric asthma hospitalization disparities persist over time with no narrowing
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